MIL-T-85236(AS)
25 August 1978

MILITARY SPECIFICATION

TEST SET, SWITCHING UNIT

TS-3634/ARM

This specification is approved for use by the Naval Air Systems Command, Department of the Navy and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. The equipment covered by this specification shall be capable of performing functional tests and to fault isolate the SA-1964/ARC-159 (V) Switching Unit to the piece part. The Test Set, Switching Unit, TS-3634/ARM hereinafter referred to as the Test Set shall be for use at the intermediate and depot levels of maintenance. The Test Set shall conform with MIL-T-28800 Type I, Class 4, Style F except as modified herein.

1.2 Classification. The equipment covered by this specification shall consist of the following item:

Type Designation	Application Paragraph
Test Set, Switching Unit, TS-3634/ARM	3.7

1.3 Associated Equipment. This equipment shall operate with the associated equipment:

Item Name	Federal Mfg. Code	Manufacturers Part No.
Audio Oscillator	28480	200CD
Digital Multimeter	89536	8800A/AA
Oscilloscope	28480	1551/20
Switching Unit	13499	SA-1964/ARC-159(V)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4920

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2. APPLICABLE DOCUMENTS

2.1 General. The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-L-3661	Lampholder, Indicator Light, Indicator-light Housings, and Indicator light Lenses, General Specifications for
MIL-S-3786	Switches, Rotary (Circuit Selector, Low Current Capacity) General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-E-17555	Electronic and Electrical Equipment, Accessories and Repair Parts, Packaging and Packing of
MIL-T-18303	Test Procedures; Preproduction Acceptance and Life for Aircraft Electrical Equipment, Format for
MIL-N-18307	Nomenclature and Identification for Electronic, Aeronautical, and Aeronautical Support Equipment including Ground Support Equipment
MIL-T-28800	Test Equipment for use with Electical and Electronic Equipment, General Specification for
MIL-C-39019	Circuit Breakers, Magnetic, Low Power, Sealed, Trip-Free, General Specification for
MIL-C-39024	Connector, Electrical, Jacks, Tip (Test Point Panel or Printed Wiring, Type) General Specification for
MIL-S-83731	Switches, Toggle, Unsealed and Sealed Toggle, General Specifications for

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STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-471	Maintainability Demonstration
MIL-STD-704	Aircraft, Electric Power, Characteristics
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MS3106	Connector, Plug, Electric, Straight
MS3107	Connector, Plug, Electric, Quick Disconnect

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Testing. This specification makes provisions for testing (see 4).

3.2 Material, parts, and processes. The materials, parts and processes shall be in accordance with MIL-T-28800.

3.2.1 Selection of parts, materials, and processes. Except as specified herein, MIL-T-28800 shall apply for the selection of parts, materials and processes.

3.2.2 Approval of nonstandard parts and materials. Approval of nonstandard parts and material shall be in accordance with MIL-T-28800.

3.3 Design and construction. The design and construction of the equipment shall be in accordance with MIL-T-28800.

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3.3.1 Total weight. The total weight of the equipment, including cables, shall be minimum consistent with good design and shall not exceed 6.0 pounds.

3.3.2 Microelectronic items. When used, microelectronic items shall be in accordance with MIL-T-28800.

3.3.3 Electron devices. Electron tubes and semiconductor devices shall be in accordance with MIL-T-28800.

3.3.4 Modules, maintenance. The electronic portions of the equipment shall be divided into maintenance modules as specified in MIL-T-28800. Modules shall normally be considered repairable.

3.3.5 Time totalizing meter. Not applicable.

3.3.6 Cables and connectors. The equipment shall provide for the use of cables and connectors in accordance with MIL-T-28800. Insofar as is practicable, all connectors on the front panel shall be mounted along the lower edge of the panel and in no case shall their location be such that the connecting cables will interfere with the operating controls.

3.3.6.1 Cabling. The test equipment shall be compatible with the equipment to be tested. It shall be the electrical equivalent of the cabling used in the normal installation of the equipment being tested.

Designation	Type	Function
P1	MS3106F22-55S	Mates with J1 of SA-1964/ ARC-159(V) Switching Unit
P2	MS3107A16-11P	Mates with 28VDC power source
J1	MIL-C-39024/10-01	Test point - Radio 2 MIC AUD IN (H)
J2	MIL-C-39024/10-01	Test point - Radio 2 UNATTEN AUD IN
J3	MIL-C-39024/10-01	Test Point - Radio 2 XMT CONT (KEY) OUT
J4	MIL-C-39024/10-01	Test point - Radio 1 MIC AUD OUT

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J5	MIL-C-39024/10-01	Test point - Radio 1 UNATTEN AUD IN
J6	MIL-C-39024/10-01	Test Point - Radio 1 XMT CONT (KEY) OUT
J7	MIL-C-39024/10-01	Test point - BYPASS AUDIO
J8	MIL-C-39024/10-01	Test point - spare
J9	MIL-C-39024/10-01	Test Point - Radio 1 XMT CONT KEY
J10	MIL-C-39024/10-03	Test point - CHASSIS GROUND
J11	MIL-C-39024/10-01	Test point - Radio 1 X-MODE CONT IN
J12	MIL-C-39024/10-01	Test Point - Radio 2 X MODE RCV AUDIO IN
J13	MIL-C-39024/10-01	Test point - Radio 2 X-MODE CONT IN
J14	MIL-C-39024/10-01	Test point - X-MODE RCV AUDIO OUT
J15	MIL-C-39024/10-01	Test Point - Radio 2 X-MODE PLAIN AUDIO OUT
J16	MIL-C-39024/10-01	Test point - Radio 2 XMT CONT KEY (2)
J17	MIL-C-39024/10-01	Test point - Radio 1 X-MODE RCV AUD IN
J18	MIL-C-39024/10-01	Test Point - Radio 2 X-MODE XMT VO AUD OUT
J19	MIL-C-39024/10-01	Test point - Radio 2 X-MODE XMT VO AUD OUT
J20	MIL-C-39024/10-01	Test point - Radio 1 REXMT CONT IN

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J21	MIL-C-39024/10-01	Test Point - Radio 2 REXMT CONT IN
J22	MIL-C-39024/10-01	Test point - Radio 1 RCV AUD 600 ohm (H)(IN)
J23	MIL-C-39024/10-01	Test point - Radio 2 RCV AUD 600 ohm(H) (IN)
J24	MIL-C-39024/10-01	Test Point - Radio 1 X-MODE RCV AUD OUT
J25	MIL-C-39024/10-01	Test point - Radio 2 X-MODE RCV AUD OUT
J26	MIL-C-39024/10-01	Test Point - Radio 1 X-MODE XMT AUD OUT
J27	MIL-C-39024/10-01	Test point - Radio 2 X-MODE XMT AUD OUT
J28	MIL-C-39024/10-01	Test point - Radio 1 REXMT AUD OUT
J29	MIL-C-39024/10-01	Test Point - Radio 2 REXMT AUD OUT
J30	MIL-C-39024/10-01	Test point - AUTO RELAY PLAIN +27.5 VDC IN
J31	MIL-C-39024/10-01	Test point - AUTO RELAY CYPHER +27.5 VDC IN
J32	MIL-C-39024/10-01	Test Point - Radio 2 XMT VO AUD (L)
J33	MIL-C-39024/10-01	Test point - spare
J34	MIL-C-39024/10-01	Test point - spare
J35	MIL-C-39024/10-01	Test Point - Radio 2 X-MODE RCV AUDIO OUT
J36	MIL-C-39024/10-01	Test point - spare

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J37	MIL-C-39024/10-01	Test point - spare
J38	MIL-C-39024/10-01	Test point - spare
J39	MIL-C-39024/10-01	Test point - spare
J40	MIL-C-39024/10-01	Test point - spare
J41	MIL-C-39024/10-01	Test point - spare
J42	MIL-C-39024/10-01	Test point - spare
J43	MIL-C-39024/10-01	Test point - spare
J44	MIL-C-39024/10-01	Test point - spare
J45	MIL-C-39024/10-01	Test point - Radio 2 RCV AUD IN (H)
J46	MIL-C-39024/10-01	Test point - Radio 1 RCV AUD IN (H)
J47	MIL-C-39024/10-01	Test Point - Radio 2 XMT VO AUD (H)
J48	MIL-C-39024/10-01	Test point - Radio 2 MIC AUDIO
J49	MIL-C-39024/10-01	Test Point - Radio 2 RCV AUD
J50	MIL-C-39024/10-01	Test point - Radio 2 XMT KEY
J51	MIL-C-39024/10-01	Test point - Radio 1 MIC AUDIO
J52	MIL-C-39024/10-01	Test Point - Radio 1 RCV AUDIO
J53	MIL-C-39024/10-01	Test point - Radio 1 RCV AUDIO
J54	MIL-C-39024/10-01	Test Point - Radio 1 XMT VO AUDIO (H)
J55	MIL-C-39024/10-03	Test point - Radio 1 XMT VO AUDIO (L)

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J56	Dual Binding Post	Mates audio signal generator to test set audio input

3.3.7 Controls. Front panel controls and layouts shall be in accordance with Figure 1. The detailed requirements of the controls shall be as specified in 3.7.1.5.

3.3.8 Human engineering. Human engineering shall be in accordance with MIL-T-28800.

3.4 Availability. The availability shall be a minimum of 99 percent when calculated using the results of the Reliability and Maintainability demonstrations of paragraphs 3.4.1 and 3.4.2 as follows:

$$\text{Availability} = \frac{\theta}{\theta + \bar{M}_{ct}}$$

Where:

θ is the observed Mean-Time-Between-Failure (MTBF) equal to the total operating time of the equipment divided by the number of failures, as defined in MIL-STD-781.

and:

$\bar{M}_{ct}$ is equal to the Mean Corrective maintenance downtime, as defined in MIL-STD-471.

3.4.1 Reliability. The contractor shall conduct a reliability program in accordance with MIL-T-28800. The program shall be approved by the procuring activity.

3.4.1.1 Operating stability. The equipment shall operate as specified for not less than 4000 hours or 12 months, whichever comes first, without scheduled maintenance, realignment, or adjustment of any control not readily accessible to the operator during normal operation.

3.4.1.2 Operating life. The equipment shall have a minimum total operating life of 10,000 hours. Parts requiring scheduled replacement due to wear during the life of the equipment and, the wearout life of such parts, shall be determined by the contractor and submitted to the procuring activity.

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3.4.1.3 Reliability in Mean-Time-Between-Failures (MTBF). The equipment, including any built-in test provisions, shall have a minimum acceptable (operating) MTBF (e_1) of 400 hours when tested as specified in 4.4.3. The equipment shall have an inherent design reliability of 2,000 hours (nominal) MTBF.

3.4.2 Maintainability. The contractor shall conduct a maintainability program in accordance with MIL-T-28800. The program plan shall be approved by the procuring activity. Built-in-test features, modular construction and packaging, and provisions for test points shall be as specified in MIL-T-28800. The program will provide appropriate input data for Integrated Logistics support requirements in accordance with MIL-T-28800 applicable to the avionic equipment to be tested by this support equipment.

3.4.2.1 Maintenance times. The equipment, including any built-in provisions, shall be designed to meet the following quantitative maintainability requirements for corrective maintenance when tested in accordance with 4.2.2.1.

(a) Mean time to repair, $\bar{M}_{ct}$ 1 hour

(b) Maximum time to repair, $M_{max\ ct}$ 2 hours (at the 90th percentile)

3.4.2.2 Maintenance manhours per maintenance action. The total maintenance manhours per maintenance action shall not exceed 5 hours when conducted in accordance with 4.2.2.1.

3.4.3 Probability of successful performance of specific test (Compatability). The probability of successful performance during a complete test (all test sequences) shall be a minimum of 99 percent when tested in accordance with the provisions of 4.2.2.2. The probability of successful performance of a complete test of the avionic equipment to be tested by this support equipment shall include: (a) Not detecting a failure in the avionic equipment to be tested, (b) indicating a nonexistant failure in the avionic equipment to be tested, (c) the probability of causing a failure in the avionic equipment to be tested, and (d) the probability that a failure in the avionic equipment under test will cause a failure in the support equipment (except for the activation of a protective device).

3.4.4 Interference control. The generation of radio interference by the equipment and the vulnerability of the equipment to electromagnetic interference shall be controlled within the limits of MIL-STD-461.

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3.4.5 Maintenance provisions and field testing. Provisions for maintenance shall be as specified in MIL-T-28800. Specific test points and test facilities shall be provided to the greatest extent practical for ease of field testing and maintenance.

3.4.6 Nomenclature and nameplates. Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with MIL-N-18307.

3.4.7 Service conditions

3.4.7.1 Altitude. The altitude requirement shall be in accordance with MIL-T-28800 for Class 4 equipment.

3.4.7.2 Shock. The shock requirement shall be in accordance with MIL-T-28800 for Class 4 equipment.

3.4.7.3 Vibration. The vibration requirement shall be in accordance with MILT-28800 for Class 4 equipment.

3.4.7.4 Temperature and Humidity. The temperature and humidity requirement shall be in accordance with MIL-T-28800 for Class 4 equipment.

3.4.7.5 Salt atmosphere. Not applicable.

3.4.7.6 Sand and dust. Not applicable.

3.4.8 Standard conditions. The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient
Altitude	Normal ground
Humidity	Room ambient
Vibration	None
Input Power	28VDC ±2 VDC

3.5 Transportability. Transportability requirements shall be in accordance with MIL-T-28800.

3.6 Performance characteristics. The performance of the equipment shall be as specified herein.

3.7 Detailed requirements.

3.7.1 Test Set, Switching Unit, TS-3634/ARM. The Test Set shall meet the following requirements.

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3.7.1.1 Function. The Test Set shall be capable of interfacing with the SA-1964/ARC-159(V) Switching Unit. The Test Set shall be capable of individually testing the Switching Unit. The Test Set shall consist of a bench type enclosed chassis unit. The Test Set shall be designed to be used with the Common Ground Support Equipment listed in paragraph 6.6.

3.7.1.2 Form factor. The Test Set outline dimensions shall not exceed those specified in Figure 2.

3.7.1.3 Weight. The weight of the Test Set including cables shall not exceed 6.0 pounds.

3.7.1.4 Content. The Test Set shall contain the following assemblies, subassemblies, circuits and cables:

(1) Test Set Panel (including integral chassis and covers)

The Test Set is a self-contained unit with test points and switches which interface and test the Switching Unit through pendant cable. Another pendant cable will provide primary power to the Test Set.

3.7.1.5 Controls

3.7.1.5.1 Test Set panel controls. The front panel controls layout and labeling shall be as shown in Figure 1.

3.7.1.5.1.1 Test Set circuit breaker CBI. Circuit breaker, CBI, shall control the primary DC power to the Test Set and to the unit under test. The circuit breaker shall be capable of interrupting both sides of the primary DC power lines. CBI shall conform to MIL-C-39019.

3.7.1.5.1.2 RADIO 1 switch S1. Switch, S1, shall be a toggle switch which will control the operating mode, PLAIN or CYPHER, of the No. 1 Radio, AN/ARC-159(V) Radio Set. This switch simulates a control function of the AN/ARC-159(V) Radio Set. S1 shall conform to MIL-S-83731.

3.7.1.5.1.3 RADIO 2 switch S2. Switch, S2, shall be a toggle switch which will control the operating mode, PLAIN or CYPHER, of the No. 2 Radio, AN/ARC-159(V) Radio Set. This switch simulates a control function of the AN/ARC-159(V) Radio Set. S2 shall conform to MIL-S-83731.

3.7.1.5.1.4 AUTO RELAY switch S3. Switch, S3, shall be a toggle switch which will simulate those functions of the AN/ARC-159(V) Radio Set which configure the Switching Unit for "automatic relay" models of operation. When S3 is in the OFF position, the Switching Unit shall be

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configured to provide a "non-automatic" operating mode. When S3 is in the PLAIN position the Switching Unit shall be configured to provide an "automatic relay" mode of operation in which normally received voice signals will be automatically relayed. When S3 is in the CYPHER position the Switching Unit shall be configured to provide an "automatic" relay mode of operation in which (secure voice) signals will be automatically relayed. Also when S3 is in the CYPHER position, the Test Set shall provide the resistance loads necessary for test of the Switching Unit. S3 shall conform to MIL-S-83731.

3.7.1.5.1.5 XMT switch S4. Switch, S4, shall be toggle switch which provides manual selection of the "transmit" mode of operation for Radios 1 and 2 of the AN/ARC-159(V) Radio Set. This switch is a three position switch providing OFF, RADIO No. 1 transmit or RADIO 2 transmit control. Switch, S4, simulates the "press-to-talk" controls. Switch S4 shall control the illumination of Test Set XMT RAD 1 and XMT RAD 2 indicator lamps. S4 shall control indicator lamps to annunciate the radio which is manually selected into the "transmit" mode of operation. Switch, S4, shall conform to MIL-S-83731.

3.7.1.5.1.6 AUDIO INPUT switch S5. Switch, S5, shall be a four position rotary switch which shall select the Switching Unit circuit to which an audio frequency signal, from a source external to the Test Set, will be applied. S5 shall conform to MIL-S-3786.

3.7.1.6 Electrical connections. Electrical connections to external circuits shall be provided as follows:

NOTE: Test Set test jack field, J1 through J55, are connected to pins of connector P1; i.e., connector P1-A shall be connected to J1, pin P1-HH shall be connected to J55, etc. The Test Set front panel shall have the test jack field marked with corresponding connector P1 pin designations on the front surface and shall have corresponding "J" numbers marked on the rear surface.

Designation	Type	Function
P1	MS3106F22-55S	Mates with J1 of SA-1964/ ARC-159(V) Switching Unit
P2	MS3107A16-11P	Mates with 28VDC power source
J1 thru J8	MIL-C-39024/10-01	Mates high side of external indicator (*) with Tested Unit circuits

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J9 and J55	MIL-C-39024/10-03	Mates low side of external indicator (*) with Tested Unit and Test Set chassis ground
J10 thru J54	MIL-C-39024/10-01	Mates high side of external indicator (*) with Tested Unit circuits
J56	Dual Binding Post	Mates with external audio signal source

(*) NOTE: Indicator may be a multimeter or oscilloscope.

3.7.1.7 Enclosure. The chassis and components within the Test Set shall be protected by an enclosure. The enclosure shall be designed such that the Test Set can rest on any of its six sides without distortion on the top cover or chassis and without damage to any Test Set components.

3.7.1.8 Cables. Cables shall be provided as follows:

(1) A cable shall be provided to interface with the SA-1964/ARC-159(V) Switching Unit. This cable shall be pendant to the Test Set and shall be 2.0 ft. ±2.0 inches in length.

(2) A cable shall be provided to interface with the shop 28VDC primary power source. This cable shall be pendent to the Test Set and shall be 6.0 ft. ±2.0 inches in length.

3.7.1.9 Indicators. The Test Set shall have the following indicators.

3.7.1.9.1 TEST SET 28VDC indicator, DS-1. Indicator, DS-1, shall illuminate whenever circuit breaker, CB1, is turned ON and power is supplied to the Test Set. DS-1 shall conform to MIL-L-3661 and shall contain a green lens.

3.7.1.9.2 TEST SET RAD 1 indicator, DS-2. Indicator, DS-2 shall be illuminated whenever the XMT switch, S4, is set to the RAD 1 position. DS-2 shall conform to MIL-L-3661 and shall contain a white lens.

3.7.1.9.3 TEST SET RAD 2 indicator, DS-3. Indicator, DS-3, shall be illuminated whenever the XMT switch, S4, is set to the RAD 2 position. DS-3 shall conform to MIL-L-3661 and shall contain a white lens.

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3.7.1.10 Electrical power source. The test equipment shall operate from a power source as specified in MIL-T-28800. The power required shall be of the following type and shall not exceed the specified amount.

DC Power, 28 ±2.0 VDC, 0.5 ampere

3.7.1.10.1 Protection. The equipment shall be capable of withstanding input power variations as imposed by abnormal electrical system operations as defined in MIL-STD-704

3.7.1.11 Warm-up time. No warm up time shall be required prior to operation.

3.8 Dimensions. The maximum overall size shall not exceed 13.2 inches wide by 5.3 inches high by 8.0 inches long not including the pendant cables.

3.9 Finish. Finishes shall be in accordance with MIL-T-28800.

3.10 Interchangeability Interchangeability shall be in accordance with MIL-T-28800.

3.11 Workmanship. Workmanship shall be in accordance with MIL-T-28800.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

1. First article (preproduction) tests (See 4.4)

2. Initial production tests (See 4.6)

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Acceptance tests

Life test

4.3 Inspection Conditions. The tests shall include all tests deemed necessary by the procuring activity, to determine that the equipment meets all the requirements of this specification, other applicable specifications and the contract. Within the limits specified in MIL-T-28800, first article tests shall include environmental tests in accordance with MIL-T-5422 and interference test in accordance with MIL-STD-462 to the MIL-STD-461 requirements.

4.4 First article (preproduction) tests. First article (preproduction) tests shall be made on an equipment representative of the production equipments to be supplied under the contract. The tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.4.1 First article (preproduction) test data. When specified in the contract first article test data shall be supplied in accordance with UDI-I-21349.

4.4.2 Maintainability assurance test. A maintainability insurance test approved by the procuring activity shall be conducted in accordance with MIL-STD-471 to demonstrate the requirements of 3.4.2. The test details shall include as a minimum, task selection, test method, demonstration support material, demonstration conditions, demonstration test team, and test schedule.

4.4.3 Probability of successful performance test. A probability of successful performance assurance test to demonstrate the requirement of 3.4.3 shall be conducted. The test procedure shall be prepared in accordance with 4.6 and submitted by the contractor to the procuring activity for approval.

4.4.4 First article (preproduction) approval. Approval of the test sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipment shall be delivered prior to the approval of the test sample. Prefabrication of production equipment prior to the approval of the test sample is at the contractor's own risk. The approved test sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The test sample shall not be considered as one of the equipments under the contract.

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4.5 Production equipment. Equipment supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved test sample. Each equipment shall be capable of successfully passing the same tests as imposed on the test sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.6 Initial production tests. One of the first ten production equipment shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. The test sample shall not be selected for this test.

4.6.1 Scope of tests. This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved test sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.6.2 Accessory material. In addition to the complete equipment submitted for Initial Production Tests, the contractor shall also submit such accessory material and data necessary to test the equipment.

4.6.3 Initial production sample approval. Approval of the initial production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the initial production sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.6.4 Reconditioning of initial production test sample. On completion of the initial production test, the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.7 Acceptance tests. The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all accept-

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able tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following.

(1)	Individual tests	(4.4.1)
(2)	Sampling tests	(4.4.2)
(3)	Reliability assurance tests	(4.4.3)
(4)	Special tests	(4.4.4)

4.7.1 Individual tests. Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1)	Examination of product	(4.4.1.1)
(2)	Operational test	(4.4.1.2)
(3)	Manufacturing run in test	(4.4.1.3)

4.7.1.1 Examination of product. Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.7.1.2 Operational test. Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation. 1/

1/ Approval of the operational test procedure must be obtained from the Procuring activity prior to testing.

4.7.1.3 Manufacturing run in test. Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

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Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 Hz (excluding resonant points) and a minimum amplitude of ±3 g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operating within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to the plane for 5 minutes. Where it is not feasible to vibrate the equipment in two directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain that the failure was not a result of the vibration. Should repetitive failures occur, corrective actions shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.7.1.4 Environmental tests

4.7.1.4.1 Altitude. The equipment shall be tested in accordance with MIL-T-28800, Class 4 equipment.

4.7.1.4.2 Shock. The equipment shall be tested in accordance with MIL-T-28800, Class 4 equipment.

4.7.1.4.3 Vibration. The equipment shall be tested in accordance with MIL-T-28800, Class 4 equipment.

4.7.1.4.4 Humidity. The equipment shall be tested in accordance with MIL-T-28800, Class 4 equipment.

4.7.2 Sampling. Equipment selected for sampling shall first have passed the individual tests. Equipments shall be selected for sampling tests by the goverment inspector in accordance with the following: (Sampling tests shall not be conducted unless Reliability Assurance Tests of 4.4.3 are deleted by contractual action).

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Quantity of Equipment Offered for acceptance	Quantity to be Selected for Sampling Tests
First 10	0 2/
Next 50	1
Next 75	1
Next 100	1
One for each additional 200 or fraction thereof.	

4.7.2.1 Scope of tests. As a minimum each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, marking all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual and special tests.

(3) Manufacturing run in test specified in paragraph 4.4.1.3 except that the test duration shall be 120 hours with no restrictions on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.7.3 Reliability assurance tests. Reliability assurance tests shall be conducted using MIL-STD-781. Tests shall be conducted as required by both the Qualification Phase and Sampling Phase.

2/ If by contract action the Initial production test (4.3) is deleted, then a Sample Test shall be conducted on one equipment from the first 10 produced.

4.7.3.1 Qualification phase. Prior to the acceptance of equipments under the contract or order, a minimum of three (3) equipments shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests". The

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maximum number of equipment to be used shall be those listed in Table 5 of MIL-STD-781. For the Qualification Phase, Test Level E shall be used. The Accept-Reject Criteria for Test Plan VI shall be used.

4.7.3.2 Reliability sampling phase tests. Samples of the equipment shall be tested as outlined in MIL-STD-781, under the section entitled "Sampling Phase of Production Reliability Tests". For the Sampling Phase Test Level E shall be used. The Accept-Reject criteria for Test Plan VI shall be used to determine the length of the tests (until an accept or reject decision is reached).

4.7.3.2.1 Lot size for sampling phase. The equipments constructed during one month shall be one lot.

4.7.3.3 Test details. The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the test etc., shall be part of the test procedures to be submitted and approved by the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests.

4.7.4 Special tests. Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.7.4.1 Special test schedule. Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity.

4.7.4.2 Scope of tests. Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.7.5 Equipment failure. Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

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(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity, for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.8 Life test. The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300-hours duration and shall be conducted on equipments that have passed the individual tests. The life test sample shall be performed under the conditions specified in 4.5.1. The life test shall be selected by the government inspector in accordance with the following. (Equipments which have successfully passed the Initial Production Test Sampling Tests, Reliability Tests, or Special Tests may be selected for life tests). When reliability tests are conducted, the life test may be omitted if, during the reliability tests, a quantity of equipment equal to, or more than, that listed below receive at least 300 hours each of test time.

Quantity of Equipment Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
1 for each additional 500 or fraction thereof	

4.8.1 Test conditions. The life test shall be conducted under the following simulated service conditions:

Temperature	Room ambient
Altitude	Normal ground (0-5000 ft.)

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Humidity	Room ambient
AC Voltage	115 ±5 volts (at lowest applicable frequency)
DC Voltage	28.0 ±2.0 volts
Vibration	None

4.8.2 Test periods. The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.8.3 Performance check. At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.8.4 Test data. The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failure, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

4.8.4.1 Failure report. In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall proposed suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

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4.8.5 Reconditioning of life test samples. An equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor shall resubmit the equipment for acceptance.

4.9 Test procedures. The procedures for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification and the contract. MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.10 Reconditioning of tested equipment. Equipment which has been subjected to acceptance and life tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.11 Presubmission testing. No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.12 Rejection and retest. Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished to the government inspector.

5. PREPARATION FOR DELIVERY

5.1 General. All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555 and MIL-STD-794. In the event the equipment is not covered in MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart of Appendix D of MIL-STD-794.

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5.2 Marking. In addition to any special marking by the contract or order, each unit package and exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The equipment shall be for intermediate and depot level of maintenance and shall be capable of performing functional tests and fault isolation procedures on the SA-1964/ARC-159(V) Switching Unit.

6.2 Performance objectives. Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of the specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in MIL-T-28800, it is desired that their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.3 Nonrepairable subassemblies. As a general rule, nonrepairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplied), the non-repairable subassembly should evidence a mean-time-to-failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.4 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

6.4.1 Title number and date of this specification

6.4.2 Selection of applicable levels of packaging and packing (See 5.1).

6.4.3 Specify first article test data in accordance with UDI-T-21349, Report, First Article (preproduction) Test, on data requirement list DD 1423 as required by paragraph 4.2.1.

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6.5 Type designation. The parentheses (*), when used in this type designation, will be deleated or replaced by either a number of letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.4.6. The complete type number shall be used on nameplates shipping records and instruction books, as applicable.

6.6 Associated equipment. The equipment shall operate with the following associated equipment or their equivalent.

Item Name	Federal Mfg. code	Manufacturers Part Number
Audio Oscillator	28480	200CD
Digital Multimeter	89536	8800A/AA
Oscilloscope	28480	1551/20
Switching Unit	13499	SA-1964/ARC-159(V)

6.7 Definitions

lbs	Pounds
J	Jack
V	Volts
DC	Direct Current
MIC	Microphone
AUD	Audio
H	High
L	Low
UNATTEN	Unattenuated
CONT	Control
XMT	Transmit
X-MODE	Auxiliary Mode

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Definitions (cont'd)

RCV	Receive
VO	Voice
REXMT	Retransmit
AUTO	Automatic
S	Switch
CB	Circuit Breaker
P	Plug
DS	Lamp
In.	Inch

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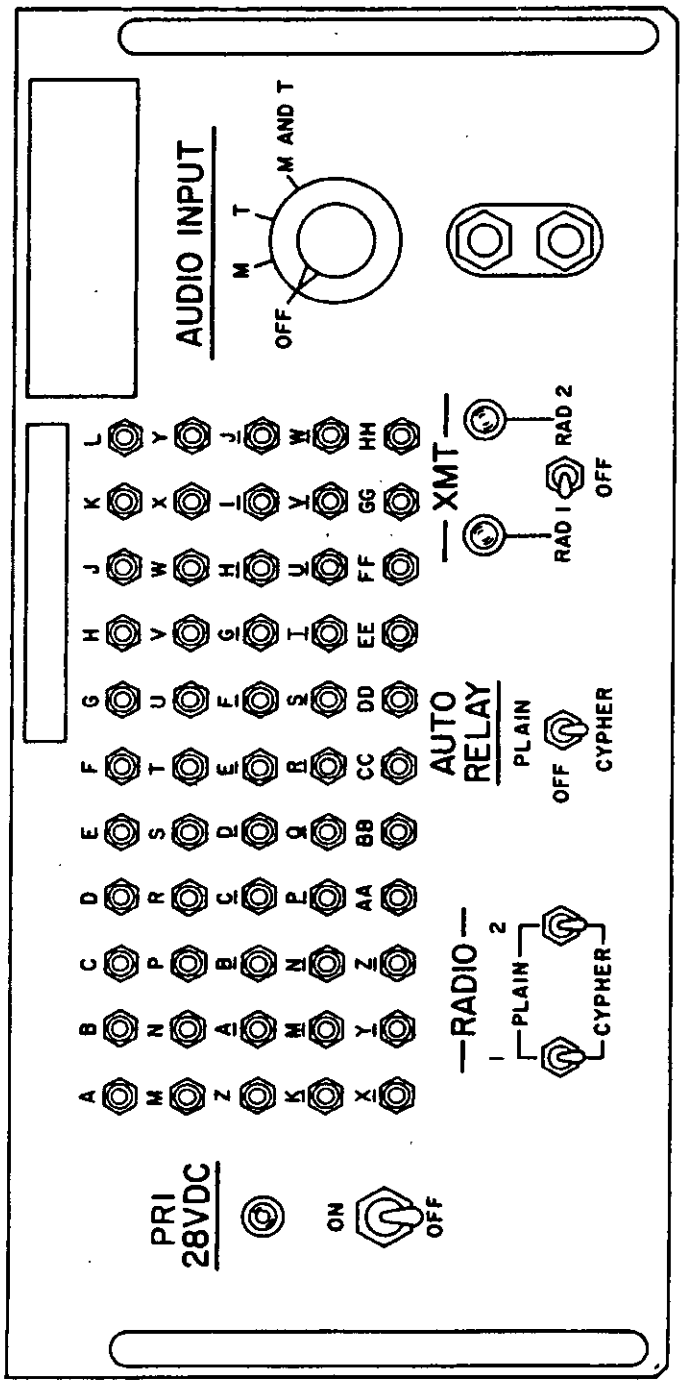


FIGURE 1
FRONT PANEL

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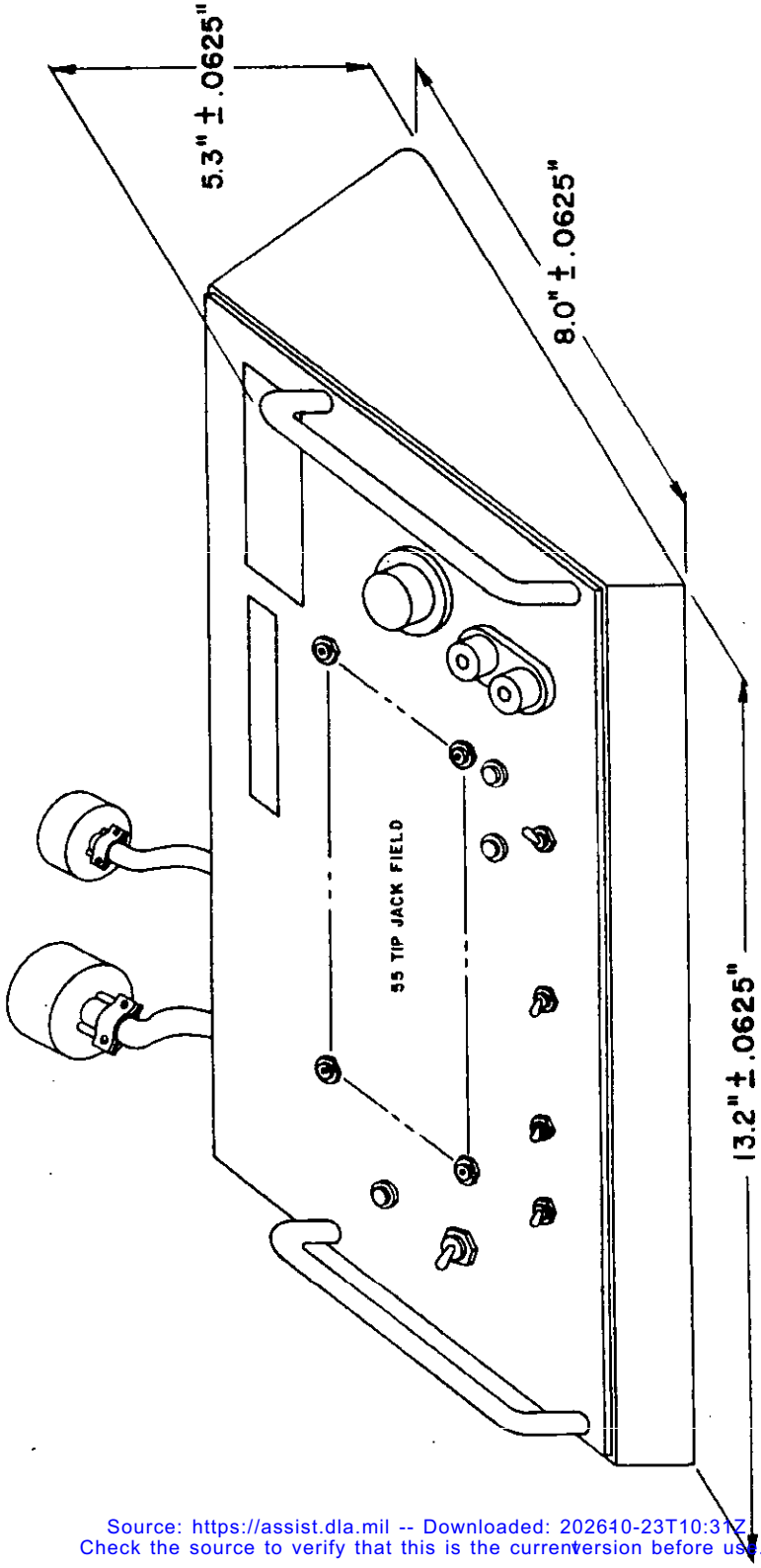


FIGURE 2
FORM FACTOR
SWITCHING UNIT TEST SET

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DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-T-85236 25 August 1978
Test Set, Switching Unit TS-3634/ARM

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

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C. REASON FOR RECOMMENDED CHANGE(S)

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